

WP-3 Resources & Availability API

Weeks: Weeks 2–3 **Track:** Backend

Goal. The endpoints that let an admin publish resources with real rules, and let a member see genuine live availability.

Tasks

- ☐ CRUD for resources (type, capacity, timezone, description) — TenantAdmin only.
- ☐ Manage availability windows per resource.
- ☐ Manage blackout periods; ensure they override availability.
- ☐ Mark resources RequiresApproval and assign approvers.
- ☐ Build an availability query: given a resource and date range, return bookable slots.
- ☐ Design clean DTOs, error contracts, and pagination.

Acceptance criteria

- ✓ An admin can publish a resource with availability and blackout rules.
- ✓ The availability query correctly excludes blackout periods and existing bookings.
- ✓ Non-admins cannot create or edit resources.
- ✓ API returns clear, structured errors.